FINCON: A Synthesized LLM Multi-Agent System with Conceptual Verbal Reinforcement for Enhanced Financial Decision Making

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Abstract

Large language models (LLMs) have shown potential in complex financial tasks, but sequential financial decision-making remains challenging due to the volatile environment and the need for intelligent risk management. While LLM-based agent systems have achieved impressive returns, optimizing multi-source information synthesis and decision-making through timely experience refinement is underexplored. We introduce FINCON, an LLM-based multi-agent framework with **CON**ceptual verbal reinforcement for diverse **FIN**ancial tasks. Inspired by real-world investment firm structures, FINCON employs a **manager-analyst hierarchy**, enabling synchronized cross-functional agent collaboration towards unified goals via natural language interactions. Its **dual-level risk-control component** enhances decision-making by monitoring daily market risk and updating systematic investment beliefs through self-critique. These **conceptualized beliefs** provide verbal reinforcement for future decisions, selectively propagated to relevant agents, improving performance while reducing unnecessary peer-to-peer communication costs. FINCON generalizes well across tasks, including single stock trading and portfolio management. [1]

1 Introduction

The intricacies and fluctuations inherent in financial markets pose significant challenges for making high-quality, sequential investment decisions. In tasks such as single stock trading and portfolio management, each intelligent decision is driven by multiple market interactions and the integration of diverse information streams, characterized by varying levels of timeliness and modalities [1, 2]. The primary objective of these tasks is to maximize profit while managing present market risks in an open-ended environment.

In practice, trading firms often depend on synthesized teamwork, structured hierarchically with functional roles such as data analysts, risk analysts, and portfolio managers communicating across levels [3, 4]. These roles are responsible for the careful integration of diverse resources. However, the cognitive limitations of human team members can hinder their capacity to rapidly process market signals and achieve optimal investment outcomes [5].

To enhance investment returns and address the limitations of human decision-making, various studies have explored methods such as deep reinforcement learning (DRL) to develop agent systems that simulate market environments and automate investment strategies [6, 7, 8]. Concurrently,

[1]We will release the code and demo in the following repo https://github.com/The-FinAI/FinCon

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advancements in large language models (LLMs) have shown great potential in performing complex tasks, including reasoning [9, 10], tool-using [11], planning [12], decision-making [13, 14], and even in various financial applications [15, 16, 17, 18, 19], suggesting they may surpass existing agent architectures. Language agents, in particular, are distinguished by their human-like communication and flexible, prompt-based structures, making them well-suited to diverse decision-making settings [20, 21, 22, 23].

To achieve optimal decision-making performance, two critical factors must be considered: (1) Organizing agents to facilitate effective teamwork and efficient communication, and (2) Enabling agents to continuously learn and refine their actions. Studies have shown that mimicking human organizational structures can successfully coordinate language agents for specific tasks [24, 25, 26]. Additionally, recent advances in textual gradient-based prompt optimization [27, 28] and verbal reinforcement [29, 30] have proven effective in iteratively improving the reasoning and decision-making capabilities of language agents.

Language agent systems designed for financial decision-making, such as FINGPT [31], FINMEM [32], and FINAGENT [33], have shown strong performance. However, they face several limitations. First, their reliance on agents' risk preferences based on short-term market fluctuations fails to control long-term risk exposure, potentially overlooking fundamental factors driving investment returns. A more effective approach is to quantify investment risks using established **measures of risk** from quantitative finance [34, 35]. Second, these systems are often limited to single-asset trading tasks, making them less adaptable to multi-asset financial applications like portfolio management. Third, they place significant pressure on a single agent to understand and process information within a constrained context window, which can degrade decision quality. Although approaches like STOCKAGENT [36] use multi-agent systems for stock trading, their reliance on extensive discussions between numerous LLM agents leads to high communication costs and slow decision-making. Moreover, the absence of a clear optimization objective can compromise outcome effectiveness. Additional related work in the literature is discussed in the Appendix A.1.

To address these issues, we propose FINCON, an LLM-based multi-agent framework for critical financial tasks, such as single-stock trading and portfolio management, as shown in Figure 1. Our main contributions are: **1)** Inspired by real-world investment roles, we introduce a novel **Synthesized Manager-Analyst hierarchical communication structure** with a risk-control component. This structure allocates financial data from different sources to corresponding functional analyst agents, allowing them to focus on specific insights, while the manager consolidates these inputs to make informed trading decisions. The streamlined communication reduces redundant peer-to-peer interaction, lowering costs and improving efficiency. **2)** Our framework generalizes beyond stock trading to handle **portfolio management**, an area not previously addressed by other financial language agent systems. **3)** We developed a **dual-level risk control component** to update risk assessments both within and across episodes. Within episodes, risk is supervised using the Conditional Value at Risk (CVaR), a quantile-based risk measure [37]. Across episodes, we introduced a **verbal reinforcement** mechanism, where investment beliefs are updated based on reasoning trajectories and profit-and-loss (PnL) trends, distilled into **conceptual perspectives**. These insights are selectively back-propagated from the manager to relevant analyst agents. Our ablation studies demonstrate the effectiveness of this risk control design in managing market risk and enhancing trading performance.

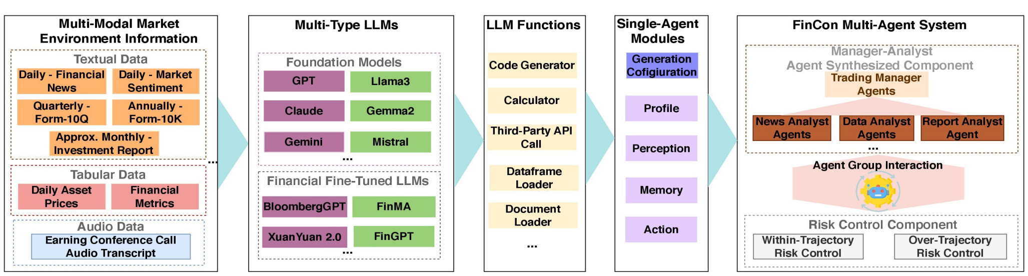

Figure 1: The general framework of FINCON.

2 Preliminaries

Here, we outline the mathematical notations for the two major financial decision-making tasks that will be explicitly discussed in our work. We also formally present the generalized modeling formulation using a Partially Observable Markov Decision Process (**POMDP**) [38] for financial decision-making tasks.

2.1 Financial Decision-making Tasks Formulation

Single Stock Trading Tasks. FINCON uses analyst agents group $\{M_{pr}^i\}_{i=1}^I$ to process multi-modal market information sources. The processed information is then used by a manager agent M_a to make trading decisions (buy, sell, hold), and to provide relevant reasoning texts. Note that the "sell" signal means the system makes a "short-selling" decision, that is, a negative trading position is allowed. Additionally, FINCON evaluates the daily investment risk, followed by prompt-optimization for the manager agent from risk-control component M_r.

Portfolio Trading Tasks. In addition to processing multi-modal market information, the analyst agents also construct a stock pool for portfolio management by considering the statistical correlations between stock returns. The manager agent then makes trading decisions for each stock in the pool. Finally, the manager agent determines the portfolio weights for all stocks using an external optimization solver that applies the mean-variance optimization described below [39]:

$$\max_{\mathbf{w}} \langle \mathbf{w}, \mu \rangle - \langle \mathbf{w}, \Sigma \mathbf{w} \rangle \quad \text{s.t. } w_n = \begin{cases} \in [0,1], & \text{"buy"} \\ \in [-1,0], & \text{"sell"} \\ = 0, & \text{"hold"} \end{cases}, \quad \forall n \in \{1, \cdots, N\} \tag{1}$$

where $\mathbf{w} = (w_1 \cdots, w_N) \in \mathbb{R}^N$ is portfolio weights vector, μ and Σ are the shrinkage estimators of N-dimensional sample expected return and $N \times N$ sample covariance matrix of chosen stocks' daily return sequences respectively [35]. We note that portfolio weights are rebalanced on daily basis. In our implementation, we begin by calculating the portfolio weights through solving the aforementioned optimization problem. Next, the target positions are determined by linearly scaling these portfolio weights from the previous step.

2.2 Modeling Quantitative Trading as POMDP

Formally, we model quantitative trading task as an infinite horizon POMDP [40, 41] with time index $\mathbb{T} = \{0, 1, 2, \cdots\}$ and discount factor $\alpha \in (0, 1]$. The components of this model are as follows: (1) a state space $\mathcal{X} \times \mathcal{Y}$ where $\mathcal{X}$ is the observable component and $\mathcal{Y}$ is unobservable component of the financial market; (2) the action space of analyst agents group is $\mathcal{A} = \prod_{i=1}^I \mathcal{A}^i$, where $\mathcal{A}^i$ represents the collection of processed market information in textual format done by agent i (total I analyst agents), and for manager agent, its action space is $\mathbb{A}$, which is modeled as $\{$*"buy", "sell", "hold"*$\}$ for single stock trading task and as $(\{$*"buy", "sell", "hold"*$\} \times [-1, 1])^{\otimes N}$ for portfolio management task among N-stocks; (3) the reward function $\mathcal{R}(o, b, a) : \mathcal{X} \times \mathcal{Y} \times \mathbb{A} \to \mathbb{R}$ uses daily profit & loss (PnL) as the output; (4) the observation process $\{O_t\}_{t\in\mathbb{T}} \subseteq \mathcal{X}$ is an I-dimensional process, with the i^{th} entry $\{O_t^i\}_{t\in\mathbb{T}}$ representing one type of uni-modal information flow solely processed by the analyst agent i; (5) the reflection process $\{B_t\}_{t\in\mathbb{T}} \subseteq \mathcal{Y}$ represents the manager agent's self-reflection, which is updated from B_t to B_{t+1} on daily basis [42]); (7) the processed information flow $\hat{O}_t = (\hat{O}_t^1, \cdots, \hat{O}_t^I) \in \mathcal{A}, \forall\, t \in \mathbb{T}$, which represents the information processing outputs from analyst agents group.

Then, our multi-agent system is supposed to learn the policies of all agents: the policies of analyst agents $\pi_{\theta^i}^i : \mathcal{X} \to \mathcal{A}^i, i \in \{1, \cdots, I\}$ (the ways to process information, i.e. $\hat{O}_t^i \sim \pi_{\theta^i}^i(\cdot|O_t^i)$), and the policy of manager agent $\pi_{\theta^a} : \mathcal{A} \times \mathcal{Y} \to \mathbb{A}$ (the ways to make trading decisions, i.e. $A_t \sim \pi_{\theta^a}(\cdot|\hat{O}_t, B_t)$) such that the system maximizes cumulative trading reward while controlling risk [43]. All policies $\Pi_{\boldsymbol{\theta}} = (\{\pi_{\theta^i}^i\}_{i=1}^I, \pi_{\theta^a})$ are parameterized by textual prompts $\boldsymbol{\theta} = (\{\theta^i\}_{i=1}^I, \theta^a)$. By updating prompts via the risk-control component M_r, the whole system optimizes policies $\Pi_{\boldsymbol{\theta}}$ in a verbal reinforcement manner. By denoting daily profit & loss (PnL) by $R_t^{\Pi_{\boldsymbol{\theta}}} = \mathcal{R}(O_t, B_t, A_t)$, the